



Ahmedabad University
School of Engineering and Applied Science

Brain Tumor Segmentation and Localization Using Traditional Image Processing and OpenCV

Weekly Status Report — Phase 1: MRI Preprocessing Week 2 of 10

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Instructor:	Professor Mehul Raval	Project No.:	20
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Team Members & Point of Contact

Name	Role	AU ID	Email
Abdulwahab Shaikh	Team Leader	AU2440099	abdulwahab.s@ahduni.edu.in
Srujan Yendugula	Member (Phase 1 Lead)	AU2440158	yendugula.s@ahduni.edu.in
Harshraj Solanki	Member	AU2440218	harshraj.s@ahduni.edu.in
Yash Kyada	Member	AU2340218	yash.k4@ahduni.edu.in

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1 Introduction

Brain tumor segmentation from Magnetic Resonance Imaging (MRI) is a critical step in neuro-oncology diagnosis and treatment planning. Raw MRI scans suffer from acquisition noise, low local contrast, and intensity non-uniformities (bias fields). Preprocessing standardizes intensity distributions and enhances tissue-tumor boundaries, forming the essential baseline required before applying segmentation algorithms.

2 Objective

The objective for this week (Phase 1) is to build and evaluate a robust, classical MRI preprocessing pipeline that converts raw 3D MRI volume inputs (BraTS FLAIR NIfTI files) into high-contrast, noise-reduced 2D axial slices ready for Phase 2 thresholding and morphological segmentation.

3 Literature Review

We utilize classical pre-processing methods and integrate them into our pipeline design, combining them together. In our three reference research papers, we used 3D Slicer [4] software for 3D medical volume handling and visualization:

1. **Non-linear median filtering for noise suppression:** Dawngliana et al. [1] have shown that median filtering is an effective noise suppression method, which is a non-linear filter. Impulse acquisition noise is effectively filtered out from brain MRI scans, while still retaining the sharp region boundaries of the anatomical and tumor areas.
2. **Histogram Equalization & Contrast Enhancement:** Murthy and Sadashivappa [2] pointed out that applying histogram equalization to each MRI slice greatly improves local contrast of tissues, which needs to be performed before global thresholding and morphological operations.
3. **FLAIR Slice Processing Pipeline:** Islam et al. [3] validated the workflow of converting 3D BraTS FLAIR volumes into 2D single-channel grayscale slices processed sequentially via median filtering and adaptive contrast enhancement (CLAHE) for CAD (Computer Aided Diagnosis) systems.

4 Work Completed This Week

4.1 Methodology

We implemented a complete 4-stage classical image processing pipeline based on the foundational steps in [1–4]:

$$\text{Raw3DNIfTI}(3DSlicer) \longrightarrow \text{GrayscaleSliceExtractor} \longrightarrow \text{MedianFilter}(3 \times 3) \longrightarrow \text{CLAHEEnhancer} \longrightarrow \text{PreprocessedMRI}$$

4.2 Implementation Details

The pipeline was developed in Python utilizing OpenCV (cv2), NumPy, Matplotlib, and 3D Slicer / NiBabel framework [4].

- **3D Volume & Slice Handling:** 3D Slicer volume orientation standards were used to extract 2D axial slices (240×240 spatial resolution) from BraTS FLAIR NIfTI volumes in 8-bit single-channel grayscale format ($[0, 255]$ intensity range).
- **Median Filtering:** Evaluated kernel sizes ($3 \times 3, 5 \times 5, 7 \times 7$) inspired by [1, 3]. A 3×3 kernel was selected as optimal because 5×5 and 7×7 kernels caused unacceptable blurring of fine tumor boundary details.
- **Contrast Limited Adaptive Histogram Equalization (CLAHE):** Grid search conducted across clip limits (1.0, 2.0, 4.0) and contextual tile grid sizes ($4 \times 4, 8 \times 8, 16 \times 16$) based on adaptive histogram principles in [2]. Selected parameters: `clipLimit = 2.0`, `tileGridSize = (8, 8)`.

5 Results

Testing across multiple representative axial FLAIR MRI slices from the BraTS2020 dataset demonstrated significant contrast improvement and noise reduction.

5.1 Visual Progression & Pipeline Stages

Figure 1 shows sequential transformations for a raw BraTS axial MRI FLAIR in grayscale. The enhancements applied include: standardization, 3×3 median filtering, and CLAHE contrast enhancement.

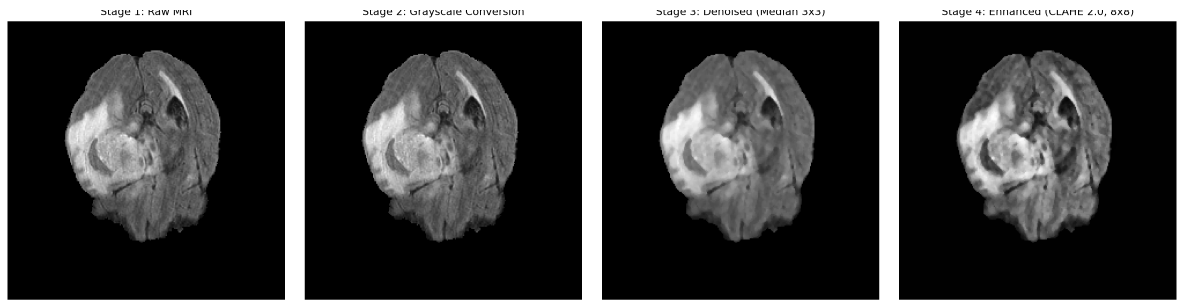


Figure 1: Visual progression across the Phase 1 MRI Preprocessing pipeline stages.

5.2 Parameter Optimization Experiments

Figure 2 presents the experimental comparison of median filter kernel sizes. The 3×3 kernel successfully suppresses salt-and-pepper acquisition noise without degrading fine anatomical edges, whereas 5×5 and 7×7 kernels introduce significant spatial blurring along tumor margins.

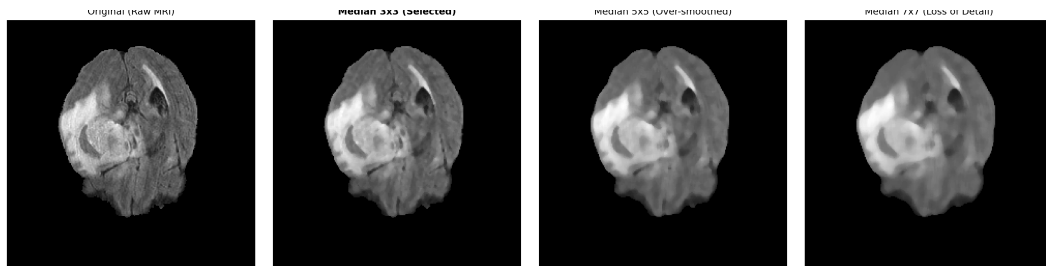


Figure 2: Experimental comparison of Median Filter kernel sizes (3×3 vs 5×5 vs 7×7).

Figure 3 displays the parameter grid search for CLAHE contrast enhancement across clip limits (1.0, 2.0, 4.0) and tile grid sizes ($4 \times 4, 8 \times 8, 16 \times 16$). Using `clipLimit = 2.0` and `tileGridSize = (8, 8)` achieved

maximum hyperintense lesion enhancement without background noise amplification, even in the presence of large hyperintense lesions.

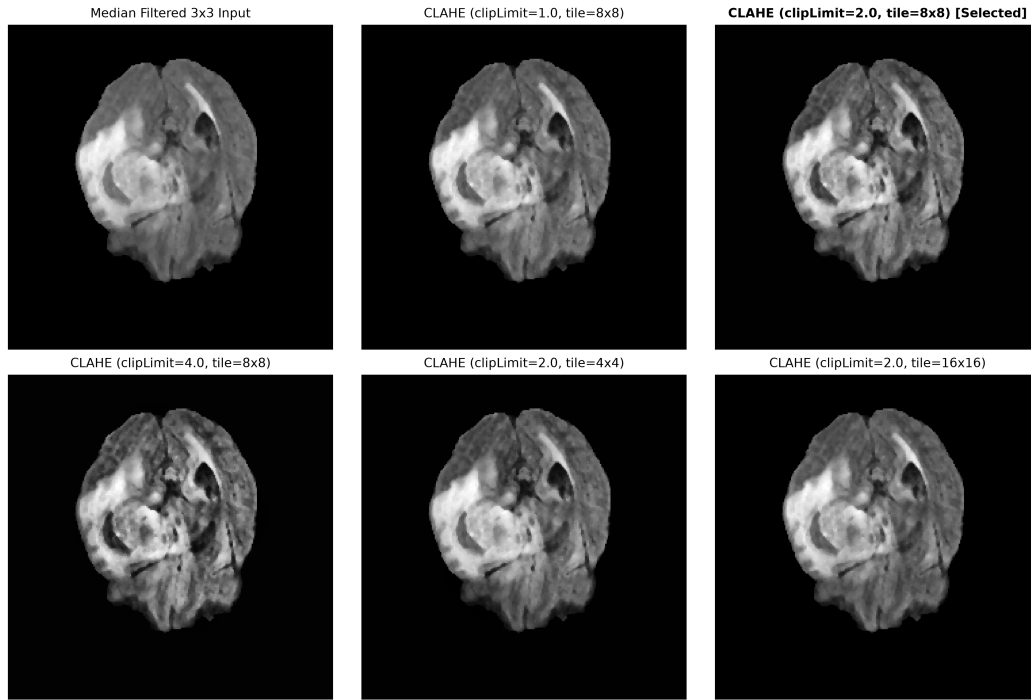


Figure 3: Parameter optimization grid for CLAHE contrast enhancement.

5.3 Quantitative Intensity Statistics

Table 1 displays the quantitative intensity metrics taken before and after preprocessing for the dataset. The real BraTS20_Training_001 FLAIR slices were reconstructed into slices along the axial axis.

Table 1: Quantitative summary of MRI intensity statistics from real BraTS20_Training_001 FLAIR dataset before and after Phase 1 Preprocessing.

Slice ID	Raw Mean $\pm$ Std	Preprocessed Mean $\pm$ Std	Min Intensity	Max Intensity
Slice 001 (Index 65)	22.50 $\pm$ 38.76	31.83 $\pm$ 50.54	0	179
Slice 002 (Index 75)	23.43 $\pm$ 39.56	32.91 $\pm$ 51.22	0	180
Slice 003 (Index 85)	21.74 $\pm$ 38.09	30.96 $\pm$ 49.24	0	179
Slice 004 (Index 95)	18.37 $\pm$ 34.02	26.75 $\pm$ 45.31	0	168
Slice 005 (Index 105)	13.48 $\pm$ 26.36	19.94 $\pm$ 34.39	0	137

5.4 Histogram Analysis

Figure 4 compares the intensity distribution histograms before and after CLAHE enhancement (excluding zero background pixels). CLAHE successfully stretches the tissue intensity distribution, broadening the gray-level spectrum and separating hyperintense tumor peaks from surrounding brain parenchymal tissue.

All cleaned slice files have been exported to preprocessed/ directory for Phase 2 integration.

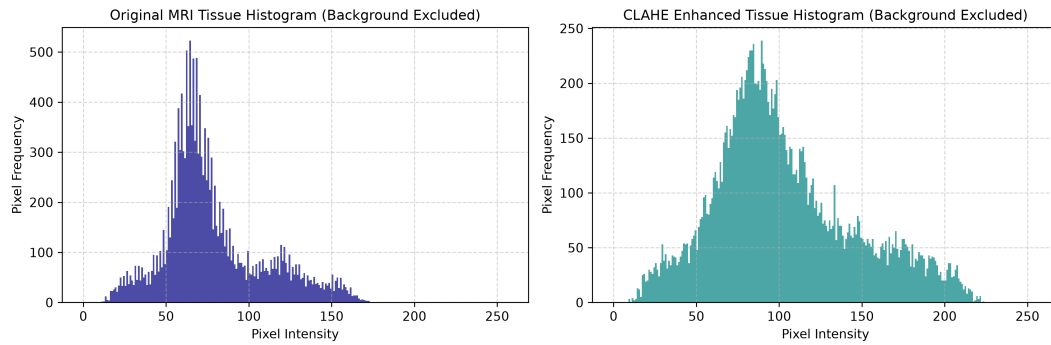


Figure 4: Comparison of MRI intensity histograms before and after CLAHE enhancement (zero background pixels excluded).

6 Challenges Faced

1. **Noise Amplification via High Clip Limits:** Higher clip limits (`clipLimit` = 4.0) amplified subtle background acquisition noise into unwanted bright speckles. Setting `clipLimit` = 2.0 prevented this artifact.
2. **Boundary Blurring:** Larger median kernels (5×5 and above) degraded fine edge definitions of small tumor extensions. Reverting to 3×3 provided optimal noise filtering while maintaining edge sharpness.

7 Plan for Next Week (Phase 2)

With the preprocessed MRI dataset ready in `preprocessed/`, Phase 2 will focus on segmentation:

PreprocessedMRI $\longrightarrow$ *Multilevel/OtsuThresholding* $\longrightarrow$ *MorphologicalFiltering* $\longrightarrow$ *TumorLocalization*

References

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- [4] A. Fedorov, R. Beichel, J. Kalpathy-Cramer, J. Finet, J.-C. Fillion-Robin, S. Pujol, C. Bauer, D. Jennings, F. Fennessy, M. Sonka, J. Buatti, S. Aylward, J. V. Miller, S. Pieper, and R. Kikinis, "3D Slicer as an image computing platform for the Quantitative Imaging Network," *Magnetic Resonance Imaging*, Vol. 30, No. 9, 2012, pp. 1323–1341.